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# MAGNETIC POLARIZATION OF SEDIMENTARY ROCKS AND THE EARTH'S MAGNETIC HISTORY

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ABSTRACT

Magnetic polarization measurements were made on 99 samples of flat-lying, undisturbed sedimentary rocks collected at eight sites in the States of Colorado, Utah, Idaho, Wyoming, Washington, and South Dakota. The majority of samples ranged in age from Pliocene to Eocene, being thus from 5 or 10 million to 50 or 60 million years old. A few samples were obtained from Jurassic formations 125 to 150 million years old. The direction of polarization in the plane of bedding, or declination, was found in two-thirds of the samples to fall in a narrow range centering on geographic rather than present geomagnetic north. Present geomagnetic north at the eight sites ranges from 13° to 22° east of geographic north. The dip or inclination of polarization in two-thirds of the samples falls in a small range centering on 63°, which compares closely with the value of 69° for the average inclination of the earth's present field at the sites sampled. The results, though few, are consistent with the idea that for 50 million years or so the polarity of the earth's magnetic field has been the same as now and the magnetic axis has had an average orientation that coincides with the earth's geographic axis.

## INTRODUCTION

A program of study of the magnetic polarization of materials of the earth's crust was begun at the Department of Terrestrial Magnetism in 1937 [see 1 of "References" at end of paper], at which time a sensitive electromagnetic method of measuring the polarization was devised [2]. Measurements were made during 1937 and 1938 of the magnetic polarization of glacial clays laid down about 20,000 years ago in the vicinity of New Haven and Hartford, Connecticut [3].

By such work, and its extension to include measurements on sedimentary rocks of great geologic age in all parts of the world, it was hoped to extend knowledge of the history of the earth's magnetism back into very ancient times. It was thought that the results would be significant if it could be shown that flat-lying, undisturbed sedimentary formations, consolidated and unconsolidated, had some kind of systematic pattern of polarization over wide regions in each geologic age, as well as from age to age. Furthermore, strong evidence for the applicability of rock polarizations to the determination of the past history of the earth's magnetic field would be obtained if it could be established that the polarizations existing now had been retained by the sediments from the time they were laid down.

## PREVIOUS RESULTS ON CLAY SEDIMENTS

In the work of 1937 and 1938, measurements on 200 years of clay deposits (laid down in annual layers, or "varves") at New Haven and 500 years of deposition at Hartford gave values of the horizontal component of polarization differing markedly from the value of magnetic declination now prevailing in that region; at New Haven the average value was found to be 30° west of present declination and at Hartford 30° east [3].

The Department's work in crustal magnetism was resumed in 1946 with further work on the clays at two localities in New England, and in 1947, with an improved high-sensitivity remanent magnetometer [4] installed in a large laboratory truck, a more extensive program was undertaken and 1,000 clay samples were obtained in the Connecticut River Valley at several localities from Hartford, Connecticut, to Inwood, Vermont [5]. These samples represented nearly 5,000 years of deposition. The results obtained for the horizontal component of polarization for these samples, taken as representing the variation in declination of the earth's field during the 5,000-year period, showed a range of values between 35° east and 25° west of present declination, much as in the early work, with the changes between these limits taking place slowly and requiring periods of 750 to 2,000 years for their completion. Inclination of the magnetic field through the 5,000 years showed a gradual change from about 45° for the oldest varves at Hartford to 65° for the youngest varves in middle Vermont; these values are to be compared with the present inclination of 75°. The difference of 20° in inclination over the 5,000-year interval and the additional 10° difference between the youngest varves and present-day inclination may not represent actual changes in geomagnetic inclination, since very preliminary laboratory tests have indicated that compaction with time and other factors of sedimentation, involving size and shape of the small particles being deposited, may account for some part, and perhaps all, of the differences. Evidence that the clays

retain the polarization acquired when laid down was obtained at two sites during the 1947 work; in Vermont a disturbed section of clay and in Connecticut a folded section both had very scattered polarizations quite unlike the systematic polarizations found in the flat-lying, undisturbed layers in these localities. Since the magnetic field existing during the ensuing 15,000 to 20,000 years had not brought these scattered polarizations into some systematic pattern, it seems quite unlikely that the same field would have altered the polarizations in the 5,000 layers of undisturbed material on which the extended series of magnetic measurements were made.

Most significantly, when the data for the folded clays were adjusted geometrically to represent conditions prevailing in the original flat-lying layers before folding, the adjusted polarizations became systematic and similar to those in undisturbed layers near-by. This aspect of the work has been given detailed treatment by Graham in a separate paper in this issue [6]. The conclusion to be drawn appeared to be that the magnetic field of the earth 20,000 years ago had about the same orientation it has today with respect to the earth's geographic axis.

To initiate the search for information concerning the field in more distant times, plans were made early in 1948 for a reconnaissance expedition during the summer months to those parts of the United States where flat-lying, sedimentary rocks, well dated by geologists, might be favorably exposed for satisfactory sampling.

## 1948 ITINERARY

From preliminary tests of rock samples taken in New York State in 1947, it was believed that most rocks would exhibit much weaker polarization than the clays. To test this point, a request was made to Dr. W. H. Bradley, Chief Geologist

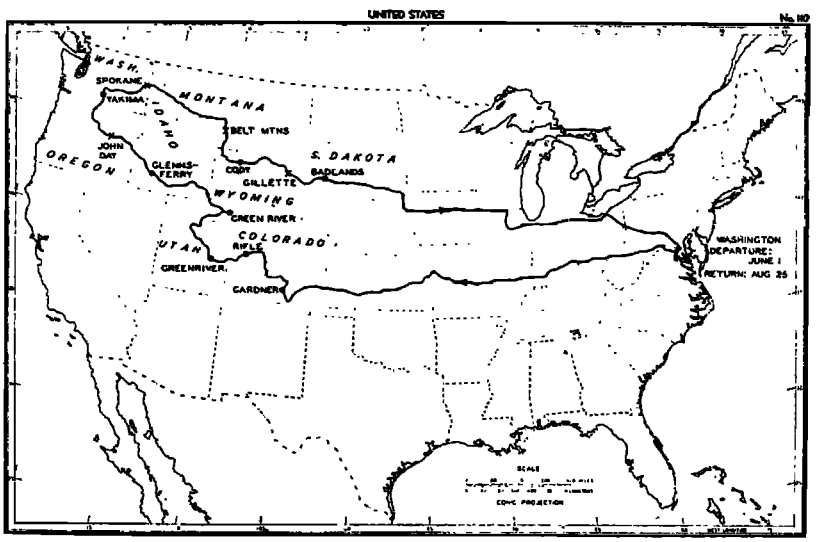



*FIG. 1—EXPEDITION FOR THE STUDY OF MAGNETIC POLARIZATIONS OF SEDIMENTARY ROCKS IN THE WESTERN UNITED STATES JUNE–AUGUST 1948*

of the U. S. Geological Survey, and his colleagues, for small samples of sedimentary rock from various parts of the United States. Prompt cooperation brought 37 specimens; 12 of these proved to be too weakly polarized to be measurable with existing equipment (moments less than $10^{-7}$ cgs/cc), while 25 were measurable. Two of these were 25 per cent as strongly polarized as the New England clays, while one was 10 per cent as strong, and the remaining 22 only 2 per cent as strong. The most strongly polarized sample had been obtained at Selah, near Yakima, Washington, and it seemed desirable, on the basis of this result, to include Selah in any proposed itinerary for the summer.

In conferences with members of the Survey during March and April, 1948, an itinerary for the expedition was arranged, restricting work to the sedimentary rocks in various basins of the western United States. Twelve working sites were decided on, in the eight States of Colorado, Utah, Idaho, Oregon, Washington, Montana, Wyoming, and South Dakota. These were to be visited in the order named, except for a trip into southwestern Wyoming after leaving Utah. The mileage from Washington, D. C., and return was estimated at 8,000 miles. This itinerary, shown in Figure 1, was strictly adhered to; eight of the twelve sites were found workable; the distance traveled was 7,750 miles in the period from June 1 to August 25.

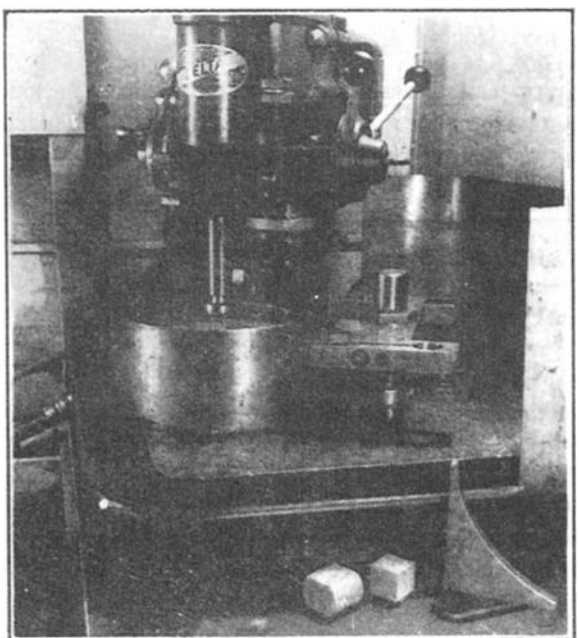


FIG. 2—TOOLS AND FIXTURES FOR CUTTING ROCK CUBES

## FIELD LABORATORY

As in the clay studies in 1947, the measuring apparatus was taken into the field in the large laboratory truck. Also installed in the truck was a drill press with

¾-H.P. motor, adapted for preparing samples of proper size and shape for use in the measuring apparatus. Diamond-impregnated coring bits and saws were provided for use with the press. With the coring bit a cylindrical sample 1-5/8 inches in diameter and 2 inches long could be obtained from a piece of rock, and then in three cutting operations, with a pair of diamond saws mounted 3 cm apart on the drill-press arbor, a 3-cm cube could be prepared. The drill press, saws, and fixture for holding the sample during cubing are shown in Figure 2.

Coolant for the coring and cubing operations was provided from a small tank containing a mixture of oil and water, the coolant being circulated by a small pump. An 80-gallon storage tank for water was also part of the truck equipment, as it was thought that water might be scarce in some localities visited, but no shortage was experienced.

The remaining major item of equipment in the truck was the 3-K.W., 60-cycle, gasoline motor-generator set, which furnished electric power whenever commercial power was not available. It was usually operated 100 to 200 feet from the truck, being lowered to the ground, when needed, by the truck hoist. By a special frequency-adjusting device added to the unit after purchase, convenient regulation of frequency could be achieved centering on 60 cycles. By loading the unit with a resistance load so as to dissipate 2 K.W. or more, the range of frequency change over long periods of operation was found to be only $\pm$ 0.2 cycle/second. For periods of the order of two or three minutes, the time usually required for a single observation, the frequency change remained within $\pm$ 0.1 cycle/second, and this variation affected the accuracy of the polarization measurements about $\pm$ 2° and the intensity measurements about $\pm$ 0.5 per cent.

## ADDITIONAL FIELD EQUIPMENT

In planning for the work on rocks in the western United States, it was anticipated that at some sites camping facilities would be required. A second large truck of the same type as the laboratory truck was fitted for sleeping, cooking, and eating, and on 21 days of the three-month expedition this provided complete living accommodations.

## THE EXPEDITION

Leaving Washington, D. C., June 1, the expedition proceeded directly to Walsenburg, Colorado, arriving there June 7, and on June 8 a working site was sought on the road to Gardner. From that date until August 18, the program outlined in the itinerary was followed, and on the latter date the return journey to Washington was begun from the Badlands, South Dakota, with arrival at destination on August 25.

From June 8 to August 18, samples were taken at 14 sites as shown in Tables 1 and 2. Table 1 summarizes the work on flat-lying undisturbed sediments, and Table 2 the work on folded sedimentary formations.

Since the major effort of the expedition was given to the study of the flat-lying undisturbed rocks, the data in Table 1 are more extensive than those in Table 2.

It will be noted from the tables that preliminary test-samples were taken at nearly every site. In this operation, which proved to be time-saving, irregularly

Table 1—Summary of polarization studies in flat-lying sedimentary rocks, June 8 to August 18, 1948

| DATES | Locality | Approximate altitude | Geologic age | Formation | Number of preliminary test samples | Number of oriented samples | Cubes | Notes |
|---|---|---|---|---|---|---|---|---|
| | | feet | | | | | | |
| June 10 | Gardner, Colorado; 2 miles west | 7,000 | Eocene | Wasatch | 2 | 5 | 5 | Very coarse-grained; unconsolidated, crumbly. Brown layers not workable; only white layers sampled. |
| June 14-17 | Anvil Pointe, (Rifle), Colorado | 8,000 | Eocene | Green River | 44 | 9 | 12 | Only a few preliminary samples sufficiently polarized to be measurable. Two cubes from oriented sample No. 3 found too weakly polarized to be measurable. One cube orientation lost. |
| June 22-24 | Greenriver, Utah; (a) San Rafel, 16-1/2 miles SW | 4,000 | Jurassic | Summerville | 8 | 6 | 7 | Ripples and wave-patterns in beds. Three cubes from oriented samples Nos. 6 and 11 too weakly polarized to be measurable. Only four cubes measured. |
| | (b) Red Rock Spring, 31 miles SW | | Jurassic | Carmel | .. | 3 | 7 | |
| June 28-July 1 | Green River, Wyoming; 1 mile west | 6,000 | Eocene | Green River (Laney Shale) | 34 | 21 | 21 | Oriented samples Nos. 15 and 16 not polarized or too weakly polarized to measure; many preliminary samples also too weak. Polarized rocks interbedded with non-polarized. |
| July 7-9 | Glenns Ferry, Idaho; 1 mile east | 2,500 | Pliocene | Payette | 13 | 13 | 17 | Unconsolidated rock; diamond tools not used for cutting. Cubes prepared by hand. Four cubes (second sample) not measured. |
| July 16-20 | Selah (Yakima), Wash. 0.5 mile west | 1,000 | Miocene | Ellensburg | 9 | 24 | 27 | Unconsolidated rock; diamond tools not used. Sedimentary beds overlie deep gravel deposit. One oriented sample regarded as slumped and discarded. |
| July 25 | Spokane, Wash.; 5 miles south | 2,000 | Miocene | Latah | 8 | .. | .. | All preliminary test samples too weakly polarized to be measurable. |
| July 27 | Garrison, Montana; 3 miles east | 4,000 | Cretaceous | ..... | .. | 4 | 4 | Flat-lying bed near fold produced during Laramide revolution. All cubes too weakly polarized to be measurable. |
| August 7-10 | Cody, Wyoming; 12 miles east | 5,000 | Eocene | Wasatch | 2 | 4 | 5 | Unconsolidated rock; material crumbly in brown layers, and samples taken from white layers. |
| August 14-18 | The Badlands, Sheep Table Mountain, South Dakota | 3,000 | Miocene | Arikaree | 7 | 21 | 40 | Unconsolidated rock; bedding shows no ripple or wave patterns in section sampled. |
| | | | Totals..... | | 127 | 110 | 145 | |
| | | | Too weak to measure or not used. | | | 11 | 16 | |
| | | | Measured..... | | | 99 | 129 | |

Table 2--Summary of polarization studies in folded sedimentary rocks, June 8 to August 18, 1948

| Dates | Locality | Approximate altitude | Geologic age | Formation | Number of preliminary test samples | Number of oriented samples | Cubes | Notes |
|---|---|---|---|---|---|---|---|---|
| | | feet | | | | | | |
| June 8, 9 | Walsenburg, Colorado; 18 miles NW | 6,500 | Cretaceous | Niobrara, Dakota sandstones; Pierre and Benton shales | 10 | .. | .. | Sandstones too weakly polarized to measure; shales too crumbly. |
| June 20, 21 | Glenwood Springs, Colo.; 4 miles west | 6,000 | Permian | Red beds | 5 | 2 | 12 | Degenerate case for stability determination. |
| June 23 | Greenriver, Utah; San Rafael Swell, 25 miles SW | 4,000 | Jurassic | Carmel sandstone | .. | 1 | 3 | Abandoned because of sampling difficulties. |
| June 30 | Green River, Wyo.; 1 mile west | 6,000 | Eocene | Laney shale, just underlying unconformity of Castle Rock sandstone | 5 | 7 | 9 | Cubes strongly polarized; samples obtained from off-shoot of an intrusive sill rather than from a fault breccia as at first believed. |
| July 16-20 | Selah (Yakima), Washington; 0.5 mile west | 1,000 | Miocene | Ellensburg (conglomerate) | .. | 13 | 14 | Five oriented samples lost in processing. 14 cubes prepared from remaining eight. Each sample removed individually from face of conglomerate exposure, after fixing orientation. Stability indicated by measurements of polarization. |
| July 27 | Garrison, Montana; 3 miles east | 4,000 | Cretaceous | Limestone | 5 | 5 | 7 | Folds only a few feet across. Samples all too weakly polarized to be measurable. |
| July 29 | Lennep, Montana; 3 miles west | 4,500 | Cretaceous | Shales and sandstone | 4 | .. | .. | Too weakly polarized to be measurable. |
| July 30 | Martinsdale, Mont.; 15 miles west on U.S. Route 6 | 4,500 | Pre-cambrian | Belt Series; limestone | 5 | .. | .. | Too weakly polarized to be measurable. |
| | | | | Totals..... | 34 | 28 | 45 | |

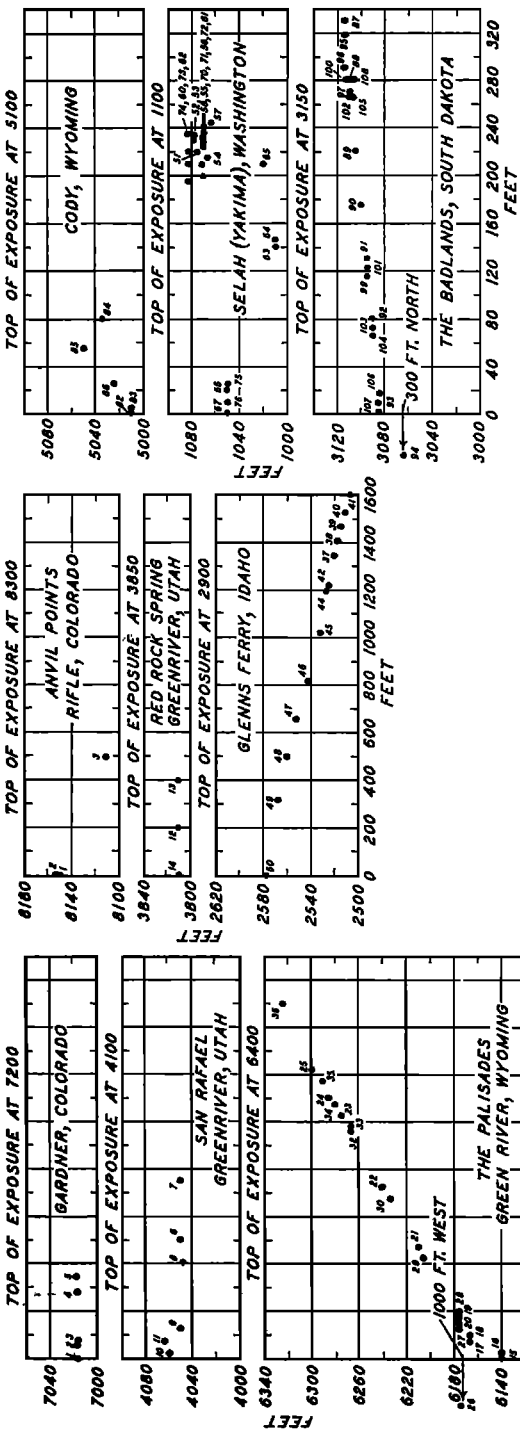



FIG. 3—LOCATIONS OF ROCK SAMPLES AT EACH WORKING SITE, JUNE-AUGUST, 1948

shaped pieces of rock, small enough to fit directly into the sample holder of the measuring apparatus, were taken at suitable spacings over the face of the exposure, without obtaining the orientations, and their intensities of magnetization measured. The spot from which each fragment was taken was conspicuously marked and numbered so that it could be found again if the preliminary sample proved sufficiently intense to warrant taking an oriented sample. By this procedure many sedimentary beds were found to be too weakly polarized to be satisfactorily measurable, and such beds were ignored in the final sampling. At Anvil Points, Colorado, and Green River, Wyoming, particularly, many beds were found unmeasurable, these being interbedded with measurably polarized layers in both cases. At Anvil Points, where the face of the exposure was examined over a vertical distance of about 600 feet, measurably polarized layers were located in only two or three feet of the whole exposure. At Green River, Wyoming, weakly polarized or non-polarized layers alternated with measurably polarized layers throughout the exposure, so that while samples for measurement were taken over a vertical extent of 150 feet or more, there were at the same time many weakly polarized or non-polarized layers in the same 150 feet.

It will be seen from the "Notes" in Tables 1 and 2 that the practice of preliminary sampling did not in every case prevent final sampling, cubing, and measuring of beds too weakly polarized to yield satisfactory data; in a few instances the entire sampling process was carried through before certain beds were recognized as being too weakly polarized.

The number of samples taken at the different sites varied greatly because of differences in accessibility. At some sites the talus slopes angled up along the face of the exposure, as at Green River, Wyoming, so that a large vertical extent as well as a considerable horizontal extent could be sampled. At other sites, such as Anvil Points, Glenns Ferry, and the Badlands, the grade of the road running at the base of the exposure permitted sampling over a large vertical expanse by working a sufficient distance along the road. For example, at the Badlands, working along 350 feet of road permitted sampling over a vertical distance of 40 feet; at Glenns Ferry, in a horizontal distance of 1,500 feet, samples were taken from a vertical distance of 80 feet. At Selah, on the other hand, where the work was done in a gravel pit, sampling opportunity was limited to a few small areas. The flat-lying sedimentary beds overlie a gravel bed 40 to 50 feet thick, the sedimentary beds being 15 to 20 feet in total thickness. Talus slopes extending above the gravel permitted sampling from two areas on the west wall of the pit; in one case across a 50-foot section of the 15- to 20-foot vertical exposure and in the other case across only a 20-foot section. Two very small areas on the east wall of the pit not far above the pit floor yielded a few samples.

The points at which oriented samples of rock were taken from the vertical or nearly vertical exposure at each site mentioned in Table 1, except Garrison, Montana, have been plotted in the various diagrams in Figure 3. Garrison was not included because no polarization data were obtained there. Views of some of the sites are given in Figure 4.

In the plot for Anvil Points in Figure 3, two points are shown at the 8,150-foot level. It should be noted that one of these represents five samples taken in a vertical

row one above the other and very close together, so that the vertical distance covered is only 10 or 12 inches. This small distance, however, probably represents many thousands of years of deposition. The second point represents three samples, also taken in a vertical row closely spaced and covering a vertical distance of only six or eight inches. All other points in the several plots represent individual samplings, the total number of samplings, including the eight just mentioned, being 106

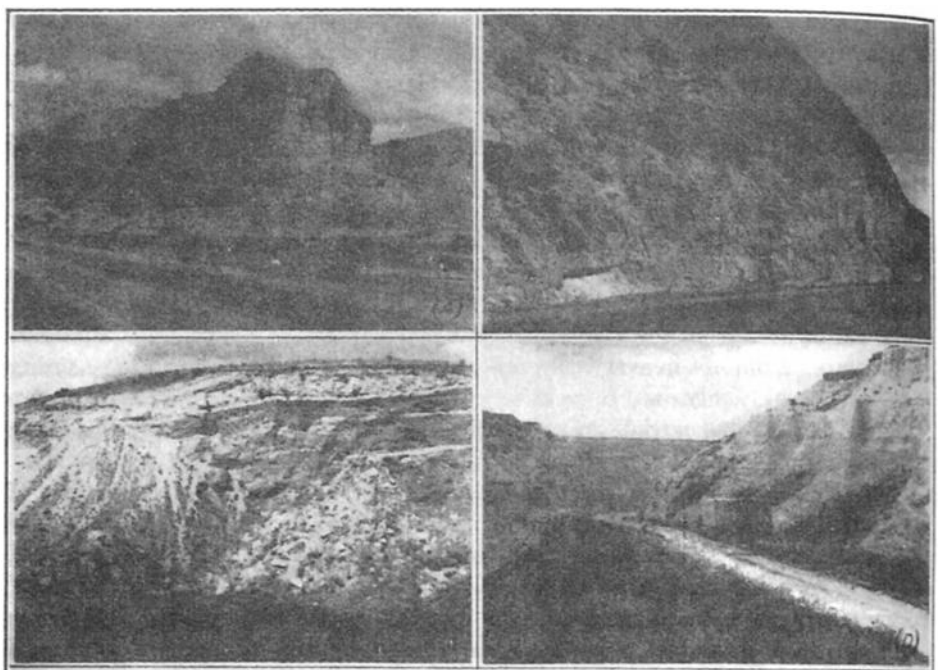


FIG. 4—VIEWS OF ROCK EXPOSURES SAMPLED JUNE TO AUGUST, 1948 (A) GREEN RIVER, WYOMING; (B) GLENNS FERRY, IDAHO; (C) SELAH, WASHINGTON; (D) THE BADLANDS, SOUTH DAKOTA

Original plans called for work on the Hagerman beds (Pliocene) near Hagerman, Idaho, but these could not be located in the course of a day's search and the exposure found at Glenns Ferry, 30 miles westward, proved to be an excellent substitute.

The Latah formation (Miocene) in the vicinity of Spokane, Washington, also listed on the original itinerary, proved to be unworkable. Exposures of this formation were found five miles south and 15 miles northwest of Spokane, 10 to 20 feet high, but overlaid by lava beds 50 or more feet thick, and penetrated by intrusives. With volcanic formations in such proximity it was concluded that the Latah formation at these sites should be classed as disturbed. Nevertheless, at the more accessible exposure of the two, that south of Spokane, eight unoriented samples were taken, all of which proved too weakly polarized to be measurable. Satisfactory exposures of the Latah formation could not be found.

The third locality found unworkable among those on the original list of sites was

that in the vicinity of the Belt Mountains and Crazy Mountains in Montana. All exposures found in this locality in two days' search were disturbed and distorted.

Between Sheridan and Gillette, Wyoming, workable sites of the Wasatch formation (Eocene) were sought from which to take samples to augment the few obtained at Cody. Suitable sites could not, however, be located.

It was found early that shales could not in most cases be sampled by methods available on the expedition. No way was developed by which to make the very thin laminations adhere to one another during coring and cubing. Accordingly, the Pierre shale seen in the vicinity of Chamberlin, South Dakota, was not sampled.

Among the exposures of folded rocks sampled, as shown in Table 2, there was none that provided satisfactory data for the study of stability of original polarization. However, the conglomerate material found at Selah, on the basis of measurements on 14 cubes cut from eight oriented samples, gave scattered polarizations very different from the polarizations found for the flat beds overlying the conglomerate, a result discussed in Graham's paper elsewhere in this issue [6].

## SAMPLING AND CUBING OPERATIONS

The chief difficulty encountered during the summer's work arose in connection with the sampling and cubing of "unconsolidated" rock. In Table 1 the notes indicate that only unconsolidated rocks were available for study at Glenns Ferry, Selah, Cody, and the Badlands. So-called "hard" rocks had been found at Rifle, Greenriver (Utah), and Green River (Wyoming), but at Glenns Ferry it was discovered that the type of rock found there could not be processed with the diamond coring bits and saws used previously on the hard rocks. The material was more like a very dry clay than a rock, and the coolant used with the diamond tools caused softening and deformation and brought about disintegration to a mud. Radically different processing was called for, as will be described later.

For the hard rocks an oriented sample usually was obtained at a place on the rock wall where a small horizontal surface, perhaps six inches by ten inches in size, could be found on which to scribe a line representing the direction of geomagnetic north. This surface might be either the top or the bottom of a rough slab projecting slightly from the main body of the rock wall, or a surface that could be readily uncovered by dislodging sections of broken rock. Slabs from three to six inches thick usually were selected. Scribing the magnetic north direction on the underside of the slab was generally more difficult than doing so on the upper side, and less accurate.

In those cases where neither top nor bottom could be exposed for marking, the vertical or nearly vertical exposed face of the selected sample was used for reference. A horizontal line was scribed on it and the direction of this line with reference to magnetic north suitably recorded, being both scratched on the rock and recorded in field notes. The "dip" also of the vertical or nearly vertical reference face was recorded.

After providing the necessary reference marks, the oriented sample was carefully broken away from the exposure with chisel and hammer, or pried loose with a crowbar. Usually several such oriented samples were collected before coring and cubing were begun.

In those cases where the orientation was established by scribing the magnetic north direction on a horizontal surface, additional lines paralleling the first would be scribed on the surface before coring, so that the cores taken, usually two or three, would each have a scribed line running across the top (or bottom) face. In the subsequent cubing operation two cube faces would be cut parallel to the scribed line, so that one "axis" of the cube would coincide with geomagnetic north. This was designated the $X$ axis. The remaining faces would then have edges in the direction of axes designated $Y$ and $Z$. A right-hand system of coordinates was used, the $+X$ direction being that of magnetic north. Arrows representing these directions were scribed on various faces of each cube for permanent record of orientation, a suitable identifying number also being scribed on each cube to indicate the exact place from which the sample was taken.

For a sample with orientation marks on a vertical or nearly vertical surface, it was necessary to transfer the horizontally scribed line from that surface to an approximately horizontal surface prepared by chipping or scraping after the sample was pried loose. Reasonably good fixtures for this purpose were devised in the field, but accuracy of orientation of the axes of the cubes prepared from such samples undoubtedly was less than that attained when the cubes were prepared from samples having the geomagnetic north direction scribed on a horizontal surface while *in situ*.

Before coring, each oriented sample was carefully blocked-up on the bed of the drill press so that the sample would be disposed both horizontally and vertically as it was *in situ* and the coring bit would cut a cylinder having its axis at right angles to the horizontal bedding of the rock.

For the unconsolidated rocks the removal of a sample from the rock exposure was handled differently from that for hard rocks. In nearly every case the hard clay-like material was chipped away above, below, and at the sides of a selected block, making a trench around the block three or four inches wide and five or six inches deep. The top surface of the block was then carefully chipped and scraped until accurately flat and horizontal, after which a line representing the direction of geomagnetic north was drawn on in pencil and the block suitably numbered. The block was then carefully detached from the rock wall by light chiseling and prying with a bar. In a number of cases the block was too badly broken up to be used when finally detached from the rock wall, and an entirely new block had to be prepared adjacent to the first one.

Preparing cylindrical cores from the unconsolidated rocks proved to be impossible, as was mentioned earlier, and cubes were fashioned by hand directly from the sample blocks. Considerable time was spent in evolving a procedure for this; brass and wooden jigs and fixtures were devised for holding the sample and guiding saws, scrapers, and files during the cubing. Eventually it was found practicable to cut the cubes crudely with hacksaws without causing contamination from the steel saw-blades, and then to prepare the final cube surfaces with sandpaper or garnet paper. No easy or rapid method for processing such cubes was improvised, and only a limited number could be taken at a site during the brief stay of a few days scheduled for each location.

During the evolution of the process of preparing samples, the effect of the tools

on the magnetic characteristics of the rocks was investigated. For example, from certain beds of unconsolidated rocks, oriented blocks were taken with both iron and brass tools, and the cubes then prepared also with iron and brass tools. No difference in magnetic direction or intensity that was considered significant was found between samples processed with the two sets of tools.

Again, in the case of consolidated rocks, oriented blocks were taken from certain beds by extensive use of hammers and chisels, and, in contrast to this, by gently prying the blocks loose. No difference in polarization was shown by samples subjected to these different treatments.

Finally, in cutting cubes of consolidated rocks with the diamond tools, the polarizations were measured just after cutting and again after polishing the cube faces with sandpaper and garnet paper. In no case was a difference in polarization noted after polishing.

Perhaps the best evidence for lack of contamination from the tools is the consistency of results, this consistency being obtained in spite of the use of a wide variety of tools by three persons, each processing the samples by different techniques and procedures in which no attention was paid to orientation of the sample.

In preparing a cube from unconsolidated rock with hacksaws and sandpaper, the originally prepared horizontal surface was generally retained as one of the cube faces. Saw-cuts were made vertically downward from this face, parallel to the line representing geomagnetic north direction and at right angles to this line to provide four more faces, after which the cube was detached from the sample block by a rough saw-cut and the sixth face smoothed off by sandpapering. Making reasonably accurate cubes by this process was laborious. Having achieved a satisfactory cube, the identifying number and the arrows representing the axes were marked on it, after which a coating of quick-drying clear lacquer was applied to prevent disintegration.

More than one cube was cut from some of the oriented samples. Table 1 shows that 145 cubes were cut from 110 oriented samples taken from flat-lying, undisturbed sedimentary formations, and Table 2 that 45 cubes were obtained from 28 samples of folded sedimentary strata or conglomerate formations.

Where more than one cube was prepared, the second cube was generally taken contiguous to the first, the two being taken side by side, not one above the other, and hence from the same rock stratum or layer. This procedure was adopted in order to provide measurements on stratigraphically identical cubes in order to obtain some indication of the accuracy of cubing and measuring.

## DETERMINATION OF DIRECTION AND INTENSITY OF POLARIZATION

To determine the direction and intensity of polarization of the samples, six measurements were made on each cube, two with reference to each of the three axes. These provided values for each of three components of direction and of intensity. From the intensity components, the total intensity $F$ was computed.

Measurements of the components of direction of polarization were plotted on an "equal-area projection" chart which, in only a minute or two, provided an angle representing the inclination of polarization from the horizontal plane and an angle for the component of polarization in the horizontal plane, the latter being the

direction taken as representing the declination of the earth's magnetic field at the time of deposition, referred to present geomagnetic north. The "equal-area projection" is described and its use discussed elsewhere in this issue [6]. Usually the three lines plotted on the chart from measurements on a cube intersected at a point, thus indicating very small error in the work of cubing and measuring. In the great majority of remaining cases, the triangle or error of the plot indicated an uncertainty of only 1° to 2° in the direction, in space, of the magnetization. Many of the samples were weakly polarized or were polarized along one of the axes of measurement, and in such cases weak components were obtained. These were difficult and tedious to measure, and the values of direction as well as intensity less accurate than those for stronger polarizations. Samples with moments of $3 \times 10^{-6}$ cgs/cc or less were in this category.

In Table 3 the measurements on 22 pairs of cubes are given to show the degree

Table 3--Test of consistency of polarization and accuracy of sampling process, using two contiguous cubes from same oriented rock sample

| Site | Sample No. | Cube No. 1 | | | Cube No. 2 | | | Differences | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | D | I | Intensity in $10^{-6}$ cgs unit/cc | D | I | Intensity in $10^{-6}$ cgs unit/cc | $D_1-D_2$ | $I_1-I_2$ | Int.$_1$-Int.$_2$ in $10^{-6}$ cgs unit/cc |
| | | ° | ° | | ° | ° | | ° | ° | |
| Anvil Points, Colo. | 1D | 342 | 60 | 254.0 | 342 | 58 | 319.0 | 0 | +2 | -65.0 |
| Greenriver, Utah | 14 | 30 | 49 | 0.6 | 10 | 77 | 0.6 | +20 | -28 | 0.0 |
| Green River, Wyo. | 29 | 355 | 63 | 0.9 | 5 | 62 | 0.9 | -10 | +1 | 0.0 |
| | 35 | 29 | 22 | 3.2 | 39 | 24 | 2.5 | -10 | -2 | +0.7 |
| Selah, Washington | 72 | 48 | 62 | 33.9 | 54 | 74 | 33.5 | -6 | -12 | +0.4 |
| Cody, Wyo. | 82 | 344 | 59 | 0.5 | 11 | 58 | 0.5 | -27 | +1 | 0.0 |
| Badlands, S. Dakota | 87 | 107 | 77 | 2.0 | 107 | 84 | 2.1 | 0 | -7 | -0.1 |
| | 90 | 141 | 24 | 2.2 | 135 | 7 | 1.7 | +6 | +17 | +0.5 |
| | 92 | 2 | 76 | 9.3 | 356 | 71 | 15.9 | +6 | +5 | -6.6 |
| | 94 | 64 | 76 | 4.8 | 64 | 72 | 3.5 | 0 | +4 | +1.3 |
| | 95 | 144 | 50 | 4.5 | 219 | 63 | 5.6 | -75 | -13 | -1.1 |
| | 97 | 45 | 60 | 6.0 | 34 | 60 | 8.3 | +11 | 0 | -2.3 |
| | 99 | 112 | -64 | 1.3 | 59 | -27 | 2.4 | +53 | -37 | -1.1 |
| | 100 | 20 | 83 | 2.0 | 18 | 67 | 1.6 | +2 | +16 | +0.4 |
| | 101 | 21 | -24 | 1.7 | 44 | -18 | 1.7 | -23 | -6 | 0.0 |
| | 102 | 359 | 77 | 1.1 | 350 | 54 | 1.5 | +9 | +23 | -0.4 |
| | 103 | 220 | 60 | 4.4 | 278 | 77 | 5.6 | -58 | -17 | -1.2 |
| | 104 | 333 | 83 | 8.8 | 356 | 70 | 14.8 | -23 | +13 | -6.0 |
| | 105 | 235 | 73 | 1.5 | 238 | 81 | 1.5 | -3 | -8 | 0.0 |
| | 106 | 173 | 19 | 2.0 | 190 | 51 | 2.5 | -17 | -32 | -0.5 |
| | 107 | 110 | 35 | 1.5 | 161 | 33 | 1.7 | -51 | +2 | -0.2 |
| | 108 | 55 | 60 | 1.4 | 18 | 65 | 1.7 | +37 | -5 | -0.3 |

to which the data for the first cube prepared agree with those of the second. Most of the data in Table 3 were obtained for cubes from the Badlands and, since a great scattering of polarizations was found for the samples at this site, it is perhaps not surprising that in a few cases even adjacent cubes show large differences in direction and intensity of polarization. In general, however, the differences between cubes in each pair range only between plus and minus 15° in values of declination, plus and minus 10° in inclination, and plus and minus $0.5 \times 10^{-6}$ cgs/cc in intensity. It should be noted that the cubes in all but three pairs of the twenty-two tabulated were prepared by hand; only the Greenriver, Utah, and Green River, Wyoming, cubes were cut with precision by the use of the diamond-impregnated tools.

To refer the declination values to geographic rather than to geomagnetic north, additive corrections were made to all values obtained by plotting, using values of present eastward declination for the western United States varying from 13° in South Dakota to 22° in Washington.

Table 4.--Measurements of magnetic polarizations of rock samples obtained in western United States

| Locality | Sample No. | D | I | Intensity in $10^{-6}$ c.g.s. per cc |
|---|---|---|---|---|
| | | ° | ° | |
| Gardner, Colorado | 1 | 19 | -90 | 2.5 |
| | 2 | 103 | -90 | 3.5 |
| | 3 | 124 | 12 | 1.8 |
| | 4 | 236 | -22 | 4.2 |
| | 5 | 266 | -50 | 2.8 |
| Anvil Points (Rifle), Colorado | 24 | 345 | 68 | 3.6 |
| | 25 | 1 | 62 | 76.6 |
| | 26 | 356 | 61 | 223.0 |
| | 1C | 342 | 69 | 162.0 |
| | 1D | 342 | 60 | 254.0 |
| | 1E | 344 | 68 | 25.4 |
| | 1F | 326 | 63 | 105.0 |
| San Rafael, Greenriver, Utah | 10 | 350 | 49 | 4.1 |
| | 7 | 216 | 13 | 0.6 |
| | 8 | 29 | 78 | 1.6 |
| | 9 | 16 | 58 | 3.2 |
| Red Rock Spring, Greenriver, Utah | 12 | 160 | 16 | 3.0 |
| | 13 | 352 | 56 | 3.6 |
| | 14 | 30 | 49 | 0.6 |
| Green River, Wyoming | 36 | 305 | 60 | 0.4 |
| | 23 | 359 | 66 | 10.5 |
| | 35 | 29 | 22 | 3.2 |
| | 24 | 332 | 71 | 3.5 |
| | 34 | 357 | 64 | 1.5 |
| | 23 | 349 | 66 | 2.9 |
| | 33 | 356 | 66 | 1.5 |
| | 32 | 356 | 67 | 1.6 |
| | 22 | 355 | 54 | 3.1 |
| | 30 | 21 | 53 | 0.4 |
| | 21 | 1 | 62 | 0.6 |
| | 29 | 355 | 63 | 0.9 |
| | 28 | 10 | 60 | 2.0 |
| | 19 | 306 | 65 | 2.7 |
| | 20 | 341 | 60 | 4.5 |
| | 27 | 4 | 66 | 2.7 |
| | 26 | 341 | 56 | 3.7 |
| | 18 | 354 | 62 | 1.1 |
| | 17 | 20 | 72 | 1.2 |
| Glenns Ferry, Idaho | 50 | 15 | 62 | 3.8 |
| | 49 | 9 | 56 | 12.0 |
| | 48 | 8 | 63 | 4.7 |
| | 47 | 357 | 63 | 15.9 |
| | 46 | 5 | 66 | 8.9 |
| | 45 | 358 | 64 | 9.6 |
| | 44 | 351 | 58 | 5.3 |
| | 42 | 355 | 64 | 8.5 |
| Glenns Ferry, Idaho (concluded) | 37 | 3 | 64 | 17.3 |
| | 38 | 357 | 56 | 11.5 |
| | 39 | 360 | 59 | 7.5 |
| | 40 | 9 | 65 | 36.0 |
| | 41 | 355 | 68 | 38.6 |
| Selah (Yakima), Washington | 74 | 360 | 44 | 65.0 |
| | 69 | 356 | 61 | 21.7 |
| | 73 | 355 | 34 | 33.1 |
| | 62 | 350 | 40 | 66.2 |
| | 52 | 11 | 64 | 86.0 |
| | 53 | 3 | 64 | 84.2 |
| | 51 | 353 | 68 | 104.0 |
| | 58 | 354 | 48 | 39.0 |
| | 55 | 8 | 62 | 100.0 |
| | 70 | 297 | 69 | 29.0 |
| | 71 | 295 | 62 | 25.3 |
| | 56 | 93 | 66 | 48.3 |
| | 72 | 48 | 62 | 33.9 |
| | 61 | 121 | 69 | 42.5 |
| | 54 | 4 | 62 | 72.4 |
| | 57 | 359 | 57 | 81.1 |
| | 67 | 188 | 50 | 1.4 |
| | 76 | 338 | 5 | 0.4 |
| | 66 | 18 | 68 | 2.1 |
| | 75 | 8 | 67 | 2.4 |
| Cody, Wyoming | 65 | 123 | -66 | 3.5 |
| | 63 | 5 | 61 | 3.5 |
| | 64 | 20 | 50 | 9.0 |
| | 85 | 213 | 72 | 0.7 |
| | 84 | 360 | 49 | 1.3 |
| | 86 | 17 | 67 | 0.6 |
| | 82 | 344 | 59 | 0.5 |
| The Badlands, South Dakota | 87 | 107 | 77 | 2.0 |
| | 95 | 144 | 50 | 4.5 |
| | 96 | 44 | 58 | 1.3 |
| | 100 | 20 | 63 | 2.0 |
| | 98 | 33 | 77 | 1.5 |
| | 97 | 45 | 60 | 6.0 |
| | 102 | 359 | 77 | 1.1 |
| | 108 | 55 | 60 | 1.4 |
| | 105 | 235 | 73 | 1.5 |
| | 89 | 13 | 6 | 3.8 |
| | 90 | 141 | 24 | 2.2 |
| | 91 | 61 | 3 | 2.3 |
| | 101 | 21 | -24 | 1.7 |
| | 99 | 112 | -64 | 1.3 |
| | 92 | 2 | 76 | 9.3 |
| | 103 | 220 | 60 | 4.4 |
| | 104 | 333 | 83 | 8.8 |
| | 106 | 179 | 19 | 2.0 |
| | 93 | 221 | 2 | 3.7 |
| | 107 | 110 | 35 | 1.5 |
| | 94 | 64 | 76 | 4.8 |

Values of declination, referred to geographic north, and of inclination, for cubes obtained from 99 oriented samples, are tabulated in Table 4, together with computed values of total intensity of polarization. The data have been arranged in Table 4

according to the vertical distribution of the samples at each site as shown in Figure 3, so that the topmost and youngest layers sampled are at the top of the group. Where several samples were taken at the same elevation, and so presumably from the same layer, or layers, the data for these appear in succession in the Table, permitting direct comparison of the results.

## DISCUSSION OF RESULTS

The values of intensity in Table 4 range from $0.4 \times 10^{-6}$ to $254.0 \times 10^{-6}$ cgs/cc; 60 per cent of the values are less than $5.0 \times 10^{-6}$ cgs/cc and 30 per cent are $2.0 \times 10^{-6}$ cgs/cc or less. In the more strongly polarized glacial clays of New England [5] the lowest values obtained were $20 \times 10^{-6}$ cgs/cc and values as high as $50 \times 10^{-6}$ to $100 \times 10^{-6}$ cgs/cc were generally found.

The samples taken at three of the eight sites represented in Table 4 were much more strongly polarized than those at the remaining five. However, it must be remembered that the samples for Anvil Points, Colorado, were taken from a very

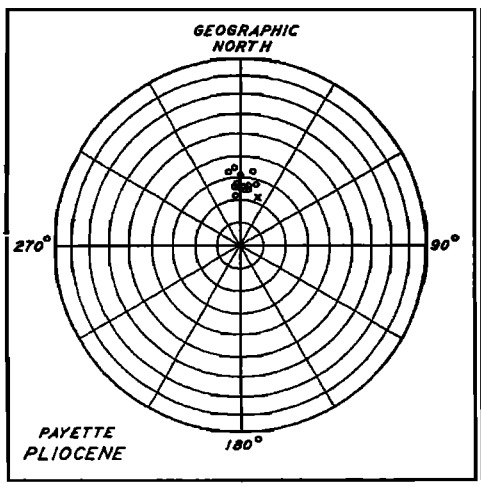



FIG. 5—EQUAL AREA PLOT OF MAGNETIC POLARIZATION IN ROCK SAMPLES, GLENNS FERRY, IDAHO
×=DIRECTION OF EARTH'S PRESENT FIELD
○=NORTH POLE

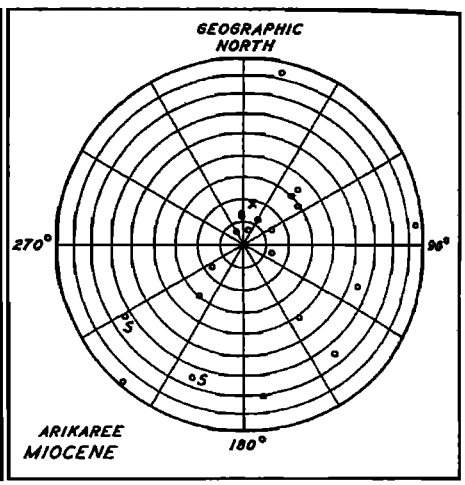



FIG. 6—EQUAL AREA PLOT OF MAGNETIC POLARIZATION IN ROCK SAMPLES, THE BADLANDS, SOUTH DAKOTA
×=DIRECTION OF EARTH'S PRESENT FIELD
○=NORTH POLE ○S=SOUTH POLE

restricted section of rock, an area about 2 feet high and less than two feet wide, and that most of the rocks above and below were too weakly polarized to be measurable, whereas the samples at Glenns Ferry and Selah were obtained over much greater areas of the respective exposures. The Glenns Ferry intensities averaged $13.8 \times 10^{-6}$ cgs/cc and the Selah values $41.5 \times 10^{-6}$ cgs/cc; the weakly polarized samples at Greenriver, Utah, Green River and Cody, Wyoming, and the Badlands, South Dakota, averaged $2.4 \times 10^{-6}$, $2.5 \times 10^{-6}$, $0.8 \times 10^{-6}$, and $3.2 \times 10^{-6}$ cgs/cc, respectively.

Inspection of the results for inclination shows that in only eight cases was the south-seeking pole downward in the samples. Four of these cases were found at

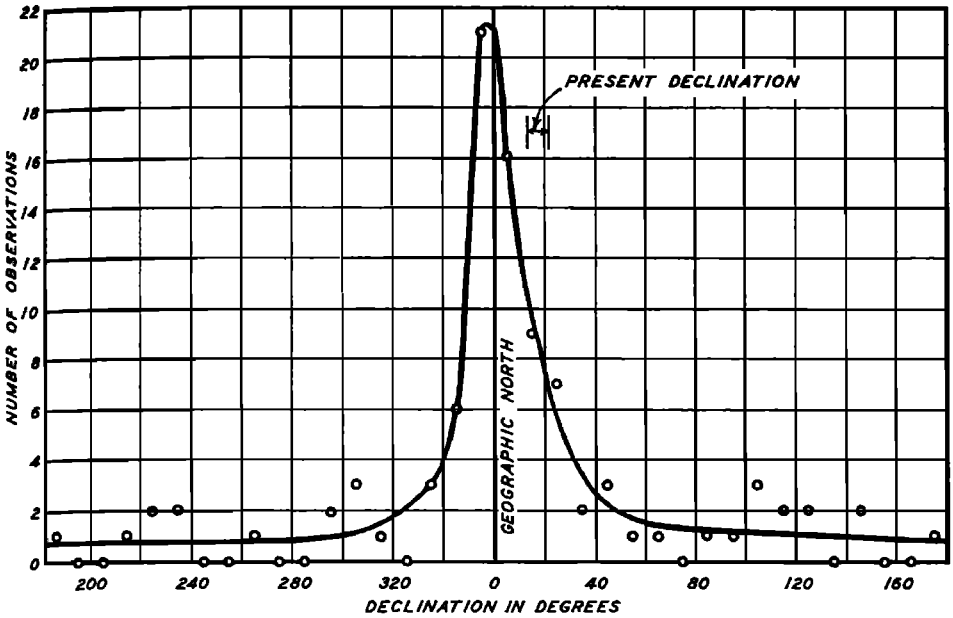



FIG. 7—FREQUENCY DISTRIBUTION OF DECLINATION MEASUREMENTS ON ROCK SAMPLES

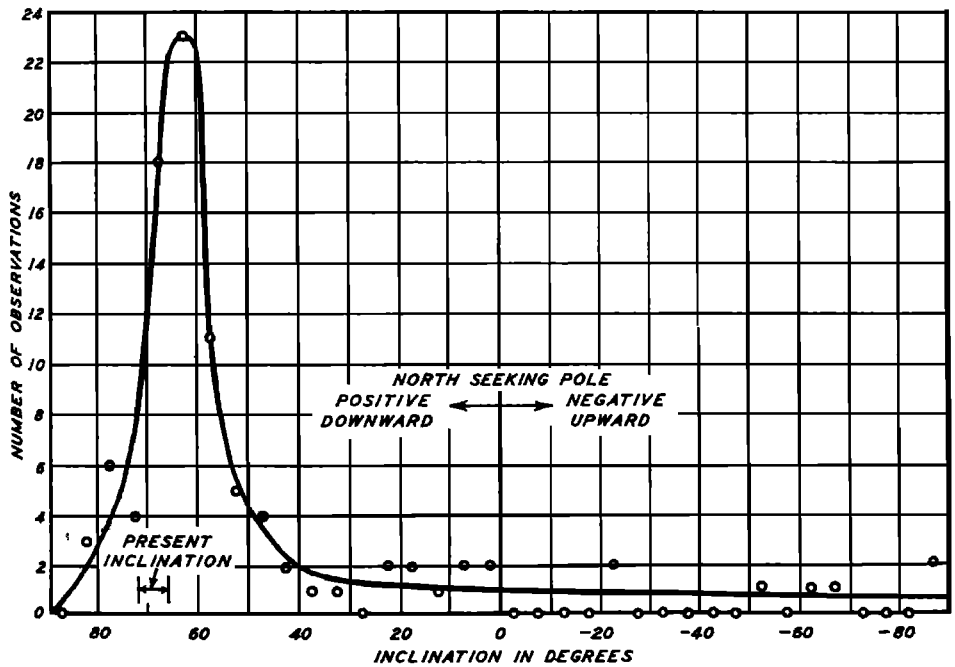



FIG. 8—FREQUENCY DISTRIBUTION OF INCLINATION MEASUREMENTS ON ROCK SAMPLES

Gardner, Colorado, two at Selah, Washington, and two in the Badlands. The Gardner results are of special interest. The exposure found there was not well suited for the polarization studies but, as it was the first locality visited and the strata were seen to be flat-lying, it was decided to undertake sampling. The unsuitable features were the coarseness of the material and its very poor consolidation. A lump of it could easily be crushed to sand by hand. Cubes could not be cut with the diamond tools, but were made by hand. Graham, in a laboratory study of the material subsequent to the expedition, has been able to demonstrate that the south-seeking magnetizations probably can be attributed to the coarseness of the particles making up the deposit.

Sample 65, one of the two taken at Selah that had negative inclination, was obtained together with samples 63 and 64 from the east side of the gravel pit, about 200 feet from the west side from which all other Selah samples were taken. The east side consisted of small lenses of sediments interbedded with gravel, being only a remnant left after the gravel excavating was stopped. The exact relation of these layers to those on the west bank was not determined.

In Figures 5 and 6 values of declination and inclination for the samples obtained at Glenns Ferry, Idaho, and the Badlands, South Dakota, are plotted. The results for Glenns Ferry show the closest agreement among polarization values for all the samples at a site, whereas the Badlands results show the greatest scatter. In the plots, the concentric circles represent 10° intervals of inclination, increasing values being measured from the periphery toward the center. Values of declination are read from geographic north at the top of the chart, clockwise around the circle through 360°. There is little question that the data in Figure 5 center on geographic north rather than geomagnetic north. Similar plots for the stations not given here show a preponderance of points in the vicinity of geographic north.

The wide scatter of polarizations in the Badlands material cannot be explained at this time. The sedimentary layers over a large area, as well as at the site worked, appear flat and undisturbed. The layers are readily distinguishable by color differences, variations in degree of weathering, and by obvious bedding planes. If, however, the material was laid down by stream flow instead of in a stagnant, or essentially stagnant lake, it may be that the scatter in polarizations resulted from such flow. The material is, however, quite fine-grained, differing very little in gross appearance from the material at Glenns Ferry in this respect. Further work at both sites should be profitable to explain both the excellent consistency of data at the one site and the scatter at the other.

To obtain the preferred values of both declination and inclination for all the results given in Table 4, the frequency-distribution curves in Figures 7 and 8 were prepared. For Figure 7 the data were averaged over 10° intervals and for Figure 8 over 5° intervals. Two-thirds of the observations of declination fall into a narrow range centering on geographic north rather than on geomagnetic north. Similarly, two-thirds of the inclination values are in a narrow range centering on 63°, a value only 6° less than the average inclination prevailing in the region sampled.

The great majority of the samples were taken from sediments laid down in the Eocene, Miocene, and Pliocene, and thus are 5 or 10 million to 50 or 60 million years old. Only a few samples were obtained from Jurassic rocks that are perhaps 125 to 150 million years old.

## CONCLUSION

The data here discussed, though representing a very small exploratory sampling of the sedimentary rocks of the earth's crust, are consistent with the idea that, for the past 50 million years or so, the polarity of the earth's magnetic field has been the same as now, and the magnetic axis has had an average orientation that coincides with the earth's geographic axis.

## ACKNOWLEDGMENTS

The work described in this paper was expedited by helpful cooperation from many sources. Mention has already been made of the planning of an itinerary with members of the U. S. Geological Survey, and their helpfulness in providing initial test samples. Much of the work of assembling these samples was done by K. E. Lohman, and we are indebted to him for this kindness.

Efforts to find a working site in the Belt and Crazy Mountains in Montana were greatly assisted by the cooperation of Mr. Tuttle of the office of the U. S. Geological Survey at Helena, Montana, who furnished maps and descriptive material for the region.

At Rifle, Colorado, we were accorded every facility at the Anvil Points Demonstration Mine, of the U. S. Bureau of Mines, by Mr. Boyd Guthrie, Engineer in Charge. Our equipment was conveniently placed for us at the mine site, and access was had to all parts of the mine for examination and sampling. Consultations with Mr. Guthrie and members of his staff provided valuable detailed information concerning the rocks being sampled.

Throughout the western areas where sites were sought and sampling was done, cordial cooperation of rangers of the U. S. Forest Service and of ranger naturalists of the U. S. Park Service was met with and was of great assistance.

Every facility for the examination of Pre-Cambrian rocks on Mt. Harrison, Idaho, was generously provided by Dr. Otto Moellmer, of Rupert, Idaho.

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